THE INDEX OF A SELF-DUAL TERNARY CODE IS DIVISIBLE BY 24: FOUR PROOFS

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Abstract. Gaiotto and Johnson-Freyd conjectured that the index of every self-dual ternary code C, the number of even minus the number of odd maximal codewords, is divisible by 24. They proved this for lengths divisible by 24 using Borcherds' theorem on theta series of even unimodular lattices, and remarked that they did not know an elementary proof. We record several proofs for all lengths. The main one is finite and elementary: a single Poisson summation over C, with two marked coordinates, shows that the maximal codewords, counted with signs, form a 2-design: $\sum_\varepsilon s(\varepsilon)\,\varepsilon_i\varepsilon_j = 0$ for $i \neq j$. Both factors 8 and 3 of 24 follow from this by short counting arguments. We also give a proof of the design identity by modular forms, a proof of the factor 8 by Kneser neighbours, and a neighbour version of the Poisson argument. The same computation with four marked coordinates gives an identity that explains why the design property stops at strength two.

1. Introduction

A ternary code of length n is a subspace $C \subseteq \mathbb{F}_3^n$. It is *self-dual* if $C = C^\perp$ for the standard form $x \cdot y = \sum_k x_k y_k$. A codeword is *maximal* if none of its entries is 0, and it is *even* or *odd* according to the parity of the number of its entries equal to 1. The *index* of a self-dual code is

$$\mathrm{Index}(C) = \#\{\text{even maximal codewords}\} - \#\{\text{odd maximal codewords}\}.$$

Gaiotto and Johnson-Freyd [1] showed that $\mathrm{Index}(C)$ is the supersymmetric (Witten) index of a holomorphic $\mathcal{N} = 1$ SVOA of central charge n attached to C, and made the following conjecture.

> **Conjecture** ([1, Conj. 1.4]; numbering of the arXiv version). The index of any self-dual ternary code is divisible by 24, even if the length is merely divisible by 12. For all lengths $12k$ there exists a self-dual ternary code of index 24.

They proved the first sentence when $24 \mid n$ [1, Thm. 1.3], and deduced $12 \mid \mathrm{Index}(C)$ when $12 \mid n$. About their proof they write: "Our proof relies on some nontrivial facts about Theta functions of lattices, and we do not know if there is an elementary proof." This note records proofs of the divisibility statement in all lengths.

Theorem. For every self-dual ternary code C of positive length, $24 \mid \mathrm{Index}(C)$.

The hypothesis $n \geq 1$ is needed: for $n = 0$ the empty word is an even maximal codeword, so Index $= 1$. It was made explicit by the formalization in Lean (Remark 8.1).

The routes are as follows.

- (0) *GJF,* $24 \mid n$ (§7): the two even neighbours of the lattice $\Lambda(C)$, and Borcherds' theorem that the Δ^k-coefficient of the theta series of an even unimodular lattice of rank $24k$ is divisible by 24.
- (A) *Kneser neighbours* (§5): a Fourier identity on $\mathbb{F}_3^n$ shows $\mathrm{Index}(C) + \mathrm{Index}(C') \equiv 0 \pmod 8$ for neighbouring codes, and the neighbour graph is connected. This gives $8 \mid$ Index. The factor 3 still comes from route (0).

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(B) *Signed 2-design, modular* (§6): the design identity (Theorem 3.1) from a theta series with harmonic coefficients and $M_2(\mathrm{SL}_2(\mathbb{Z})) = 0$. Proposition 3.2 then gives 8 and 3.
(C) *Signed 2-design, finite* (§4): the design identity from one Poisson summation over C with two marked coordinates, and again Proposition 3.2. This proof is elementary and self-contained. A neighbour version (C′) is in Remark 4.5.

Routes (B) and (C) are each a complete proof of the Theorem, and so is (A) together with (0). A comparison is in §8.

2. Preliminaries

Throughout, $n \geq 1$.

Lemma 2.1. *Let $C \subseteq \mathbb{F}_3^n$ be self-dual.*
(1) $4 \mid n$.
(2) *Every u in a self-orthogonal code $D \subseteq D^\perp$ has* $\mathrm{wt}(u) \equiv 0 \pmod 3$.
(3) *If C has a maximal codeword, then* $12 \mid n$.

Proof. (1) C is a totally isotropic subspace of half dimension of the standard quadratic space $\mathbb{F}_3^n$, so $\mathbb{F}_3^n$ is an orthogonal sum of $n/2$ hyperbolic planes. Comparing discriminants in $\mathbb{F}_3^\times/(\mathbb{F}_3^\times)^2 = \{\pm 1\}$ gives $1 = (-1)^{n/2}$, so $n/2$ is even. (2) $u \cdot u = \sum_k u_k^2 = \mathrm{wt}(u)$ in $\mathbb{F}_3$, since $a^2 = 1$ for $a \neq 0$. (3) A maximal codeword has weight n, so $3 \mid n$ by (2), and $4 \mid n$ by (1). □

Signs and lifts. Let $\varphi\colon \mathbb{F}_3 \to \{0, \pm 1\}$ be $\varphi(0) = 0$, $\varphi(1) = -1$, $\varphi(2) = 1$, and $f(x) = \prod_k \varphi(x_k)$. Then $f(x) = (-1)^{\#\{k : x_k = 1\}}$ for maximal x and $f(x) = 0$ otherwise, so $\mathrm{Index}(C) = \sum_{x \in C} f(x)$.

For a maximal x let $\varepsilon(x) \in \{\pm 1\}^n$ be its lift, $\varepsilon_k = -\varphi(x_k)$; so $\varepsilon_k = 1$ iff $x_k = 1$, and $\varepsilon(x) \equiv x \pmod 3$ coordinatewise. Put $s(\varepsilon) = (-1)^{\#\{k : \varepsilon_k = 1\}} = f(x)$, and let $\mathcal{F} = \mathcal{F}(C) \subset \{\pm 1\}^n$ be the set of lifts of the maximal codewords. Then

- $\mathrm{Index}(C) = \sum_{\varepsilon \in \mathcal{F}} s(\varepsilon)$;
- $\mathcal{F} = -\mathcal{F}$ and $s(-\varepsilon) = s(\varepsilon)$, since n is even;
- $\varepsilon \cdot \varepsilon' \equiv 0 \pmod 3$ for $\varepsilon, \varepsilon' \in \mathcal{F}$ (integer dot product), because it reduces to the $\mathbb{F}_3$-product of two codewords.

Fourier analysis on $\mathbb{F}_3^n$. Let $\omega = e^{2\pi i/3}$ and $\sqrt{-3} := \omega - \omega^2 = i\sqrt{3}$. For $h\colon \mathbb{F}_3^n \to \mathbb{C}$ put $\hat{h}(u) = \sum_x h(x)\, \omega^{u \cdot x}$.

Lemma 2.2. (1) (Poisson.) *For every linear code $D \subseteq \mathbb{F}_3^n$,* $\displaystyle\sum_{x \in D^\perp} h(x) = |D|^{-1} \sum_{u \in D} \hat{h}(u)$.
(2) *If $h(x) = \prod_k h_k(x_k)$, then $\hat{h}(u) = \prod_k \hat{h}_k(u_k)$.*
(3) *$\hat{\varphi} = \sqrt{-3} \cdot \varphi$, and the indicator $\iota = [\cdot \neq 0]$ has $\hat{\iota}(b) = 3[b = 0] - 1$. Consequently $\hat{f} = (\sqrt{-3})^n f = 3^{n/2} f$ when $4 \mid n$.*

Proof. (1) $\sum_{u \in D} \hat{h}(u) = \sum_x h(x) \sum_{u \in D} \omega^{u \cdot x} = |D| \sum_{x \in D^\perp} h(x)$. (2) is clear. (3) $\hat{\varphi}(b) = -\omega^b + \omega^{2b}$ is 0, $\omega^2 - \omega = -\sqrt{-3}$, $\omega - \omega^2 = \sqrt{-3}$ for $b = 0, 1, 2$. And $\hat{\iota}(b) = \omega^b + \omega^{2b}$ is 2 for $b = 0$ and -1 otherwise. Finally $(\sqrt{-3})^2 = -3$. □

3. The signed 2-design and why it suffices

Theorem 3.1 (signed 2-design)**.** *For every self-dual ternary code C and all coordinates i, j,*

$$\sum_{\varepsilon \in \mathcal{F}(C)} s(\varepsilon)\, \varepsilon_i \varepsilon_j = \mathrm{Index}(C)\, \delta_{ij}.$$

Equivalently, $\sum_{\varepsilon \in \mathcal{F}} s(\varepsilon)(\varepsilon \cdot \mu)^2 = \mathrm{Index}(C)\, |\mu|^2$ for all $\mu \in \mathbb{R}^n$.

The diagonal case $i = j$ is trivial, since $\varepsilon_i^2 = 1$. Proofs of the off-diagonal case are in §4 and §6.

Proposition 3.2. *Theorem 3.1 implies* $8 \mid \mathrm{Index}(C)$ *and* $3 \mid \mathrm{Index}(C)$.

Proof. If C has no maximal codeword, $\mathrm{Index}(C) = 0$. Otherwise $12 \mid n$ by Lemma 2.1.

The factor 8. Fix distinct coordinates a, b, c. For $\varepsilon \in \{\pm 1\}^n$, $[\varepsilon_a = \varepsilon_b = \varepsilon_c] = \frac{1}{4}(1 + \varepsilon_a\varepsilon_b + \varepsilon_a\varepsilon_c + \varepsilon_b\varepsilon_c)$. Multiplying by $s(\varepsilon)$, summing over $\mathcal{F}$ and using Theorem 3.1,

$$\sum_{\varepsilon\in\mathcal{F},\ \varepsilon_a=\varepsilon_b=\varepsilon_c} s(\varepsilon) = \tfrac{1}{4}\,\mathrm{Index}(C).$$

The set summed over is stable under $\varepsilon \mapsto -\varepsilon$, which has no fixed points and preserves s. So the left side is even, and $8 \mid \mathrm{Index}(C)$.

The factor 3. Fix $\varepsilon^0 \in \mathcal{F}$. For every $x \in \mathbb{Z}^n$ with $x \equiv \varepsilon^0 \pmod 3$ and every $\varepsilon \in \mathcal{F}$ we have $\varepsilon \cdot x \equiv \varepsilon \cdot \varepsilon^0 \equiv 0 \pmod 3$, so $\sum_{\mathcal{F}} s(\varepsilon)(\varepsilon \cdot x)^2 \in 9\mathbb{Z}$. By Theorem 3.1 this sum is $\mathrm{Index}(C)\,|x|^2$. Take $x = \varepsilon^0$, with $|x|^2 = n$, and $x' = \varepsilon^0 + 3\varepsilon_1^0 e_1$, with $|x'|^2 = n - 1 + 16 = n + 15$. Then 9 divides both $\mathrm{Index}(C)\,n$ and $\mathrm{Index}(C)(n+15)$, hence $9 \mid 15\,\mathrm{Index}(C)$, and $3 \mid \mathrm{Index}(C)$. □

Remark 3.3. For the extended ternary Golay code the 24 maximal codewords are all even, and Theorem 3.1 says that the 12 of them taken up to sign are the rows of a Hadamard matrix of order 12. This is the classical relation between the Golay code and the Hadamard matrix of order 12; see e.g. [5] for Hadamard matrices formed by codewords of ternary self-dual codes. Theorem 3.1 is a signed version of this property that holds for every self-dual ternary code.

4. Route C: a finite proof of the design identity

Fix distinct coordinates i, j and define

$$g_{ij}(x) = \prod_{k\neq i,j} \varphi(x_k) \cdot [x_i \neq 0]\,[x_j \neq 0], \qquad G_{ij}(X) = \sum_{x\in X} g_{ij}(x).$$

For a maximal x with lift ε we have $s(\varepsilon)\varepsilon_i\varepsilon_j = f(x)\varphi(x_i)\varphi(x_j) = g_{ij}(x)$, since $\varphi(x_i)^2 = \varphi(x_j)^2 = 1$. And g_{ij} vanishes off the maximal vectors. Hence

$$G_{ij}(C) = \sum_{\varepsilon\in\mathcal{F}(C)} s(\varepsilon)\,\varepsilon_i\varepsilon_j.$$

Lemma 4.1. $\hat{g}_{ij}(u) = (\sqrt{-3})^{\,n-2} \prod_{k\neq i,j} \varphi(u_k) \cdot \big(3[u_i = 0] - 1\big)\big(3[u_j = 0] - 1\big)$.

Proof. Lemma 2.2(2) and (3), with $h_k = \varphi$ for $k \neq i, j$ and $h_i = h_j = \iota$. □

Lemma 4.2 (pointwise identity)**.** *Suppose* $3 \mid n$ *and* $D \subseteq \mathbb{F}_3^n$ *is self-orthogonal. Then* $\hat{g}_{ij}(u) = (\sqrt{-3})^{n-2} g_{ij}(u)$ *for every* $u \in D$.

Proof. If $u_k = 0$ for some $k \neq i, j$, both sides vanish. Otherwise $\mathrm{wt}(u) = n - 2 + [u_i \neq 0] + [u_j \neq 0]$. By Lemma 2.1(2), $\mathrm{wt}(u) \equiv 0 \pmod 3$; as $3 \mid n$, the weights $n - 1$ and $n - 2$ are excluded, so $u_i \neq 0$ and $u_j \neq 0$. Then the last two factors in Lemma 4.1 are $(-1)(-1) = 1$, and $\prod_{k\neq i,j} \varphi(u_k) = g_{ij}(u)$. □

Theorem 4.3. *For every self-dual ternary code* C *and* $i \neq j$, $G_{ij}(C) = 0$. *Hence Theorem 3.1 holds, and by Proposition 3.2,* $24 \mid \mathrm{Index}(C)$.

Proof. If $3 \nmid n$, C has no maximal codewords and $G_{ij}(C) = 0$. Let $3 \mid n$. Poisson summation (Lemma 2.2(1)) with $D = C = C^\perp$, $|C| = 3^{n/2}$, and Lemma 4.2 give

$$G_{ij}(C) = 3^{-n/2} \sum_{u \in C} \hat{g}_{ij}(u) = 3^{-n/2} (\sqrt{-3})^{n-2}\, G_{ij}(C) = \pm \tfrac{1}{3}\, G_{ij}(C).$$

Hence $G_{ij}(C) = 0$. □

Remark 4.4 (what the proof uses)**.** Only three facts enter: Poisson summation over $C = C^\perp$; the one-variable transforms of φ and ι; and that the weights of self-orthogonal codes are divisible by 3. The sign of $(\sqrt{-3})^{n-2}$ is irrelevant, so the proof does not even use $4 \mid n$ (for $4 \mid n$ the factor is $-\frac{1}{3}$). In contrast, f itself is a Fourier eigenfunction with eigenvalue $3^{n/2} = |C|$, so Poisson summation over C only returns $\mathrm{Index}(C) = \mathrm{Index}(C)$. Replacing φ by ι at two coordinates multiplies the eigenvalue by $-\frac{1}{3}$ on every word a self-orthogonal code can contain, and 0 is the only fixed point of $G \mapsto -G/3$.

Remark 4.5 (route C′: the neighbour form)**.** Let $D \subseteq C$ have dimension $n/2 - 1$, so D is self-orthogonal and $|D| = 3^{n/2-1}$. Lemma 4.2 and Poisson summation over D give $G_{ij}(D^\perp) = -G_{ij}(D)$ when $4 \mid n$ and $3 \mid n$. If C' is the other self-dual code containing D, the coset decomposition in the proof of Lemma 5.2 (which only uses that g_{ij} is supported on maximal, hence isotropic, vectors) gives $G_{ij}(D^\perp) = G_{ij}(C) + G_{ij}(C') - G_{ij}(D)$. Hence $G_{ij}(C) + G_{ij}(C') = 0$. Connectivity of the neighbour graph (Lemma 5.4) and the base code $T^{n/4}$, which has no maximal codewords, give a second proof of Theorem 4.3. Unlike $G_{ij}(C)$, the value $G_{ij}(D)$ is usually nonzero, so this identity is not vacuous.

Remark 4.6 (four marked coordinates)**.** Let $K = \{a, b, c, d\}$ be four distinct coordinates, and g_K the analogue of g_{ij} with ι at the four points of K. Then $\sum_{x \in C} g_K(x) = \sum_{\mathcal{F}} s(\varepsilon)\varepsilon_a\varepsilon_b\varepsilon_c\varepsilon_d =: Q_K(C)$. In the analogue of Lemma 4.2 a codeword that is nonzero off K can now have 0 or 3 zeros in K (weights n or $n-3$), and a word with three zeros in K contributes the factor $2^3 \cdot (-1) = -8$. With $(\sqrt{-3})^{n-4} = 3^{n/2-2}$ for $4 \mid n$, Poisson summation gives $Q_K = \frac{1}{9}(Q_K - 8W_K)$, that is,

$$Q_K(C) = -W_K(C), \qquad W_K(C) = \sum_{\substack{u \in C,\ u_k \neq 0\ (k \notin K) \\ \#\{k \in K :\, u_k = 0\} = 3}} \prod_{k \notin K} \varphi(u_k).$$

So the weight-$(n-3)$ codewords whose zeros lie in K are exactly the obstruction to a signed 4-design. For the Golay code $Q_K = \pm 8 \neq 0$ for all 495 sets K, so the signed maximal words are in general not a 4-design, in line with $M_4(\mathrm{SL}_2(\mathbb{Z})) \neq 0$ in route B.

Remark 4.7 (relation to the literature)**.** Theorem 4.3 is an evaluation of a MacWilliams-type identity for the complete weight enumerator of C split at the two coordinates i, j, i.e. for a complete version of Ozeki's Jacobi polynomials [3, 4]. We have not found the design identity, or its consequences for the index, in the literature; this search was not systematic. The observation that invariant theory of the unsplit complete weight enumerator alone only gives Index $\in 6\mathbb{Z}$ or $2\mathbb{Z}$ (project notes, §3(e)) is consistent with the need for two marked coordinates.

5. Route A: Kneser neighbours and the factor 8

Throughout, $12 \mid n$ (otherwise all indices vanish). For $X \subseteq \mathbb{F}_3^n$ write $F(X) = \sum_{x \in X} f(x)$, so $F(C) = \mathrm{Index}(C)$.

Lemma 5.1. *If $D \subseteq \mathbb{F}_3^n$ is linear of dimension $n/2 - 1$, then $F(D^\perp) = 3F(D)$.*

Proof. Poisson summation and $\hat{f} = 3^{n/2} f$ (Lemma 2.2(3), using $4 \mid n$) give $F(D^\perp) = 3^{-(n/2-1)} \cdot 3^{n/2} F(D)$. □

Lemma 5.2 (neighbour identity)**.** *Let $C \neq C'$ be self-dual with $D = C \cap C'$ of dimension $n/2-1$. Then* $\mathrm{Index}(C) + \mathrm{Index}(C') = 4F(D)$.

Proof. Since $D \subseteq C = C^\perp$ and $D \subseteq C'$, we have $C + C' \subseteq D^\perp$. Both have dimension $n/2+1$, so $D^\perp = C + C'$. Pick $c \in C \setminus D$ and $c' \in C' \setminus D$. Then $c \cdot c' \neq 0$: otherwise $c' \perp D + \langle c \rangle = C$, so $c' \in C$. The nine cosets $D + ac + bc'$ $(a, b \in \mathbb{F}_3)$ partition $D^\perp$. For $ab \neq 0$ every vector x of the coset has $x \cdot x = 2ab\,(c \cdot c') \neq 0$, while maximal vectors have $x \cdot x = n = 0$ in $\mathbb{F}_3$. So these four cosets contribute nothing, and $F(D^\perp) = F(C) + F(C') - F(D)$. Lemma 5.1 finishes the proof. □

Lemma 5.3. *$F(D)$ is even for every linear D.*

Proof. $f(0) = 0$, and $x \mapsto -x$ is a fixed-point-free involution of $D \setminus \{0\}$ with $f(-x) = (-1)^n f(x) = f(x)$. □

Lemma 5.4 (connectivity)**.** *Any two self-dual codes of length n are joined by a chain of neighbours.*

Proof. Let $C_1 \neq C_2$ and pick $x \in C_2 \setminus C_1$. Since $x \notin C_1 = C_1^\perp$, the code $C_1 \cap x^\perp$ has dimension $n/2 - 1$. As x is isotropic and orthogonal to it, $C' = (C_1 \cap x^\perp) + \langle x \rangle$ is self-dual, and it is a neighbour of C_1. Moreover $C_1 \cap C_2 \subseteq C_1 \cap x^\perp$, so $C' \cap C_2 \supseteq (C_1 \cap C_2) + \langle x \rangle$, which is strictly larger than $C_1 \cap C_2$. Induct on $n/2 - \dim(C_1 \cap C_2)$. □

Theorem 5.5. $8 \mid \mathrm{Index}(C)$ *for every self-dual ternary code C.*

Proof. By Lemmas 5.2 and 5.3, $\mathrm{Index}(C') \equiv -\mathrm{Index}(C) \pmod 8$ for neighbours. The code $T^{n/4}$, a direct sum of tetracodes, has index 0: every nonzero tetracode word has weight $3 < 4$, so no word of the sum is maximal. Lemma 5.4 finishes the proof. □

With GJF's $12 \mid \mathrm{Index}(C)$ (route 0) this proves the Theorem. The same neighbour technique also shows that the numbers of even and of odd maximal codewords are each divisible by 8 (project notes, Proposition 5.5).

6. Route B: the design identity from modular forms (sketch)

Let $\Lambda = \Lambda(C) = \{x/\sqrt{3} : x \in \mathbb{Z}^n,\ x \bmod 3 \in C\}$, an odd unimodular lattice [1, §2]; $\chi = \sqrt{3}(1, \ldots, 1) \in \Lambda$ is characteristic. On the shadow $S = \Lambda + \chi/2$ put $s(\lambda + \chi/2) = (-1)^{\langle \lambda, \chi \rangle}$. Every $v \in S$ is $v = y/(2\sqrt{3})$ with all y_k odd and $y \bmod 3 \in C$, so $v^2 = |y|^2/12 \geq n/12$, with equality iff $y \in \{\pm 1\}^n$. These minimal vectors correspond to the maximal codewords, with $y = -\varepsilon$ and matching signs.

For $i \neq j$ the polynomial $H(v) = v_i v_j$ is harmonic of degree 2. Put

$$\Phi(\tau) = \sum_{v \in S} s(v) H(v)\, q^{v^2/2}.$$

Then

- $\Phi(\tau + 1) = e(n/24)\Phi(\tau)$, because $|y|^2 \equiv n \pmod{24}$ for $v \in S$;
- $\Phi(-1/\tau) = (\tau/i)^{n/2} \tau^2\, \Phi(\tau)$, by Poisson summation for theta series with harmonic coefficients (see e.g. [6]) and unimodularity, as for the Ramond–Ramond theta series in [1, Prop. 2.4].

These are the multipliers of $\eta^n \cdot \tau^2$, so Φ/η^n is a modular form of weight 2 for $\mathrm{SL}_2(\mathbb{Z})$, holomorphic at ∞ because $\Phi = O(q^{n/24})$. Since $M_2(\mathrm{SL}_2(\mathbb{Z})) = 0$ [7, Ch. VII], $\Phi = 0$. Its $q^{n/24}$-coefficient is $\frac{1}{12} \sum_{\mathcal{F}} s(\varepsilon) \varepsilon_i \varepsilon_j$, which proves Theorem 3.1.

Remark 6.1 (lattice form)**.** The argument never uses the code, only Λ. For every odd unimodular lattice L of rank $n = 12k$ whose shadow has minimum $\geq n/12$, a variant of this proof shows that the signed count of shadow vectors of norm $n/12$ is divisible by 24 (project notes, §2). Route C has no known lattice analogue.

7. Route 0: Gaiotto–Johnson-Freyd for $24 \mid n$

For $n = 24k$ let $\Lambda^{\pm}$ be the two even unimodular neighbours of $\Lambda(C)$. Then $\mathrm{Index}(C) = \Delta^{-k}(\Theta_{\Lambda^+} - \Theta_{\Lambda^-}) = a_0^+ - a_0^-$, where $a_0^{\pm}$ is the Δ^k-coefficient of $\Theta_{\Lambda^{\pm}}$ in the basis $\Delta^{k-i}c_4^{3i}$. Borcherds [2, Thm. 12.1] shows $24 \mid a_0^{\pm}$. For $12 \mid n$ this gives $24 \mid \mathrm{Index}(C \oplus C) = \mathrm{Index}(C)^2$, hence $12 \mid \mathrm{Index}(C)$ [1, §1].

8. Comparison

Route	Proves	Main input	Scope
0 (GJF)	$24 \mid$ Index if $24 \mid n$; $12 \mid$ Index	Borcherds' theorem	codes, via lattices
A	$8 \mid$ Index	Fourier on $\mathbb{F}_3^n$, neighbours	codes
B	design, hence 8 and 3	harmonic theta series, $M_2 = 0$	odd unimodular lattices
C	design, hence 8 and 3	one Poisson summation on $\mathbb{F}_3^n$	codes
C′	design, hence 8 and 3	Poisson, neighbours	codes

Route C is the shortest and the only one that is both complete and elementary. Route B is the most general. Route A gives the extra parity statement mentioned after Theorem 5.5.

Remark 8.1 (formal verification)**.** Route C together with Proposition 3.2 is formalized in Lean 4 with Mathlib (repository file `lean/GJF/Divisibility.lean`, about 660 lines). The formal statement is

$$\texttt{index_dvd_24}: \quad C = C^{\perp} \text{ in } \mathbb{F}_3^n,\ n > 0 \implies 24 \mid \#\{\text{even maximal}\} - \#\{\text{odd maximal}\},$$

with codes as $\mathbb{F}_3$-submodules of $\mathbb{F}_3^n$ and the index defined literally as in [1]. It uses only the standard axioms (propositional extensionality, choice, quotient soundness), and all finite case checks are done by the kernel (`decide`, no `native_decide`). The Fourier analysis is carried out with $\omega = e^{2\pi i/3} \in \mathbb{C}$; $|C|^2 = 3^n$ is obtained by Poisson summation of the constant function, so no dimension theory is needed. A companion file checks by `decide` that the tetracode is self-dual, so the hypotheses are satisfiable.

Remark 8.2 (verification)**.** Routes A and C, Proposition 3.2 and Remark 4.6 were checked by exact computation (scripts in the project repository); route B only through its conclusion, the design identity. Route C: the one-coordinate transforms exactly in $\mathbb{Z}[\omega]$; the pointwise identity of Lemma 4.2 by brute force in $\mathbb{Z}[\omega]$ on every codeword of three codes of length 12; $G_{ij}(C) = 0$ on random codes of lengths 12 and 24; the neighbour form on random neighbour pairs; for 40 random non-self-orthogonal codes of length 12 the correction terms are nonzero and the conclusion fails in 28 cases; and Remark 4.6 on all 495 four-sets for two codes of length 12 and on 50 four-sets at length 24. The design identity and both steps of Proposition 3.2 hold exactly on random codes of lengths 24 to 60, on the Pless symmetry codes $P(17)$, $P(23)$, $P(29)$ and on the extended quadratic-residue codes XQR(47), XQR(59); for example $|\mathrm{Index}(P(17))| = 744$ and $|\mathrm{Index}(P(29))| = 21696$.

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